

Cyntexa Revision Notes - Questions and Solutions

Two Sum

```
nums = [2,7,11,15]
target = 9

n = len(nums)

for i in range(n):
    for j in range(i+1,n):
        if nums[i] + nums[j] == target:
            print([i,j])

TC = O(n^2)
SC = O(1)
```

Equilibrium Index

```
arr = [1,7,3,6,5,6]

for i in range(len(arr)):
    left_sum = sum(arr[:i])
    right_sum = sum(arr[i+1:])

    if left_sum == right_sum:
        print(i)

TC = O(n^2)
SC = O(1)
```

Find Duplicate Element

```
arr = [1,2,3,2,4]

for i in range(len(arr)):
    for j in range(i+1, len(arr)):
        if arr[i] == arr[j]:
            print(arr[i])

TC = O(n^2)
SC = O(1)
```

Most Frequent Element

```
arr = [1,2,2,3,4,2]

max_count = 0
most_freq = arr[0]

for i in range(len(arr)):
    count = 0
    for j in range(len(arr)):
        if arr[i] == arr[j]:
            count += 1

    if count > max_count:
        max_count = count
        most_freq = arr[i]

print(most_freq)

TC = O(n^2)
SC = O(1)
```

Unique Element

```
arr = [1,2,2,3,3,4]

for i in range(len(arr)):
    count = 0

    for j in range(len(arr)):
        if arr[i] == arr[j]:
            count += 1

    if count == 1:
        print(arr[i])

TC = O(n^2)
SC = O(1)
```

Reverse String

```
s = "hello"

print(s[::-1])

TC = O(n)
SC = O(n)
```

Reverse Array

```
arr = [1,2,3,4,5]

left = 0
right = len(arr)-1

while left < right:
    arr[left], arr[right] = arr[right], arr[left]
    left += 1
    right -= 1

print(arr)

TC = O(n)
SC = O(1)
```

Factorial

```
n = 5
fact = 1

for i in range(1,n+1):
    fact *= i

print(fact)

TC = O(n)
SC = O(1)
```

Fibonacci

```
n = 7

a = 0
b = 1

for i in range(n):
    print(a,end=" ")
    c = a + b
    a = b
    b = c

TC = O(n)
```

SC = $O(1)$

Pattern - Same Number

```
1
2 * 2
3 * 3 * 3
4 * 4 * 4 * 4
```

Pattern - Pyramid

```
for i in range(1, n + 1):
    print(" " * (n - i), "*" * (2 * i - 1))
```

Pattern - Square

```
for i in range(5):
    print(" " * 5)
```

Pattern - Hollow Square

```
n = 5
for i in range(n):
    for j in range(n):
        if i == 0 or i == n-1 or j == 0 or j == n-1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```